

Roshan Raj

CV abridged

Upto Sept 2024

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Research interests

Reconstruction and Foundations of Quantum Theory, Quantum Field Theory, Quantum to Classical Transition, Gravity and EM Unification, Mathematical Physics.

Educational Details

- 2024: 5 years Integrated M.Sc. (Physics), National Institute of Technology(NIT), Surat, India. CGPA: **9.37/10**
- 2018: XII: Intermediate (Physics, Maths), Loyola High School, (CBSE), Patna, India. Marks: **77.2%**
- 2016: X: Matriculation, St. Paul's High School, (ICSE), Patna, India. Marks: **91.6%**

Master's Dissertation

Title: *Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum*
Supervisor: Dr. Vikash K Ojha, NIT-Surat; [Jul, 2023-May, 2024]–URL: shorturl.at/1Nug7

Brief Abstract: A new form of the relation between relativistic kinetic energy K and relativistic momentum $\mathbf{p}$, namely $|\mathbf{p}|^2 = (1 + \gamma)mK$, involving Lorentz factor γ is obtained. It leads to a relativistic quantum wave equation. The reported relation replaces both non-relativistic and relativistic total energy-momentum relation ($E^2 = p^2c^2 + m^2c^4$), and that has sought application in both quantum and classical relativistic mechanics. Its application predicted the contractions of Bohr radius at relativistic speeds, corrected the expression of the Relativistic Lagrangian of a free particle, etc. It has valid non-relativistic limits, at $\gamma = 1$.

Institute Mini-projects

- Title: New light on the concepts of Observable and indeterminacy in the quantum realm
Guide: Dr Vikash K Ojha, NIT-Surat– UG Project III, [Jul-Dec, 2022]–URL: shorturl.at/6lMIS
Research Theme: Quantum to Classical Transition
- Title: On the investigation of two non-neutral static bodies
Guide: Dr Sharad K Yadav, NIT-Surat– UG Project II, [Jan-May, 2022]– DOI: [10.48550/arXiv.2209.10641](https://doi.org/10.48550/arXiv.2209.10641)
Research Theme: Gravity and EM Unification
- Title: Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle
Guide: Dr. Himanshu Pandey, NIT-Surat– UG Project I, [Jul-Dec, 2021]–DOI: [10.31219/osf.io/ukx9t](https://doi.org/10.31219/osf.io/ukx9t)
Research Theme: Gravity and EM Unification

Independent Works

- Raj, Roshan. 2023. “Obtaining the Klein-Gordon Wave Equation Without Using the Quantum Operators.” OSF Preprints. DOI: [10.31219/osf.io/6hmyf](https://doi.org/10.31219/osf.io/6hmyf).
Research Theme: Reconstruction and Foundations of Quantum Theory
- Raj, Roshan. 2023. “Reviewing Observables in Classical and Quantum Mechanics.” OSF Preprints. DOI: [10.31219/osf.io/p2ufx](https://doi.org/10.31219/osf.io/p2ufx).
Research Theme: Foundations of Quantum Theory
- Raj, Roshan. 2023. “An Open-ended Story on Quantization.” OSF Preprints. DOI: [10.31219/osf.io/92bvp](https://doi.org/10.31219/osf.io/92bvp).
Research Theme: Foundations of Quantum Theory

Internship

- Title: Pramatrik Gatiki ke Chitra [*The pictures of quantum dynamics* (Hindi)]
Guide: Dr Vikash K Ojha, NIT-Surat, [May-Jul, 2022]–URL: shorturl.at/lvEIE
Research Theme: Physics Education for Hindi Readers.

Achievements

- 09/16: NTS Scholarship of merit, Government of India, New Delhi.
02/20: Winner (Physics), InQuest 4.0 Researchathon, SCOSH [Student Chapter], NIT Surat
03/23: Qualified, Graduate Aptitude Test in Engineering (GATE)

Skill Sets

- Programme: C, Python, MATLAB, Octave
Softwares: L^AT_EX, MS Excel, Mathematica, and use of GitHub.
Web Design: Basics of HTML5, CSS3
Languages: English (Intermediate), Hindi(Native)

Workshops & Schools

- 08-09/24: PFW Bangladesh: *School On Quantum Field Theory*, ICTP Italy, Et al.
07/21: *Quantum Fields, Geometry & Representation Theory*, ICTS-TIFR, Bangalore.
12/20: National workshop on *Data analysis using MS Excel*, BBD NITM, Lucknow.
08/20: National workshop on *MATLAB Tools And Applications*, BBD NITM, Lucknow.

Courses

- University (Physics): Quantum Field Theory I, Many-Body Physics and Relativistic Quantum Mechanics, Particle Physics, Nuclear Physics, Special Relativity, Quantum Mechanics I & II, Computational Physics(Python, Monte-Carlo), Computational Methods (MATLAB/Octave), Atomic and Molecular Physics, Electrodynamics, Statistical Mechanics, Astrophysics, Density Functional Theory, and Classical Mechanics.
- University (Mathematics): Singular perturbation theory, Special Functions, Linear Algebra, Tensors, Vector Calculus, Integral transforms, Real & Complex analysis, Computational Methods, Infinite Series, Solutions to ODEs and PDEs, Numerical methods, Interpolation, Partial Differentiation, solutions to linear equations systems etc.
- Audit-courses:
- Algebraic Topology in Physics, Physics Latam, Prof. Pavel Putrov (ICTP) [05-06/24]
 - A Short Course in Math History, Prof. A. Viramani (IITGn) [08-09/21]
 - Classical Electromagnetism, Prof. HC Verma (IITK) [08-12/20]
 - Learning Physics through Simple Experiments, Prof. HC Verma (IITK) [03-06/20]
 - Advanced course on Special Theory of Relativity, Prof. HC Verma (IITK) [01-05/20]
 - Basics of Quantum Mechanics, Prof. HC Verma (IITK) [08-11/19]
 - The basics of Special Theory of Relativity, Prof. HC Verma (IITK) [01-03/19]

Talk

- 13/11/21: *Unification in Mathematics*, InterAct Seminar series, AMHD, NIT-Surat

Responsibilities & Positions

- 11/22-01/23: Design-Infra Coordination Head & Member: *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, NIT Surat
- 10/21-07/22: Joint Academic Affairs Secretary: *Academic Affairs Council* NIT, Surat
- 06/20-07/22 Co-convener (03/21) & Member: *Society for Cultivation of Science and Humanities* (SCOSH) Student Chapter, NIT Surat

Other

I have always been an enthusiastic, independent seeker of truth. I will listen to mathematical beauty, ancient history, and philosophical ideas and stand up for social justice. My hobbies include Poetry (composed two collections as मनोविघ्न (MaNoVighn) and कालसंघर्ष (KālSanGharsha)) and writing articles on (Pratilipi).